

Converting Repeating Decimals to Fractions

MA.7.NSO.1.2

I can write any repeating decimal as a fraction, using algebra.

Yesterday you turned fractions into decimals.

Today you go the other way — and it takes algebra.

Warm-Up

On your own. Four minutes.

Predict, then check

- 1.** Write $\frac{4}{9}$, $\frac{5}{9}$ and $\frac{8}{9}$ as decimals. You did this yesterday.
- 2.** Predict: what fraction is $0.777\ldots$? Say how you decided.
- 3.** Predict: what fraction is $0.222\ldots$?

Notes

I. Multiplying by 10 slides the tail without changing it

- A. Multiplying by 10 moves the decimal point one place right.
- B. **If $x = 0.666\ldots$ then $10x = 6.666\ldots$ Both tails are still 666...**
- C. **Two numbers with the same tail, subtracted, wipe the tail out.**

Notes

II. Four steps, and only step 2 changes

- A. **Let x equal the repeating decimal.**
- B. **Multiply x by a power of 10 so the tails line up.**
- C. **Subtract. The repeating part disappears.**
- D. **Solve for x , then simplify.**

Notes

III. How many digits repeat decides the multiplier

- A. Count the digits in the repeating block. That is how many places you must move.
- B. **One digit repeats $\rightarrow 10x$. Two digits repeat $\rightarrow 100x$.**
- C. One digit gives a 9 underneath. Two digits give a 99.

Notes

IV. Digits before the repeat need a multiplier too

- A. Count the digits that come BEFORE the repeating block.
- B. **0.4444... → subtract x from $10x$.**
- C. **0.4555... → subtract $10x$ from $100x$.**
- D. Subtract the two that have the SAME tail. It is not always x .

Example 1

One digit repeats, nothing in front of it.

Write $0.666\ldots$ as a fraction.

Which power of 10, and why? Answer out loud.

Example 1

The four steps.

$$\text{Let } x = 0.666\dots$$

$$10x = 6.666\dots \quad \text{one digit repeats, so } 10x$$

$$10x - x = 6.666\dots - 0.666\dots$$

$$9x = 6$$

$$x = 6/9 = 2/3$$

Example 1

Check it.

$$2 \div 3 = 0.666\dots$$

Example 2

Two digits repeat.

Write $0.363636\dots$ as a fraction.

Which power of 10 this time?

Example 2

The four steps again.

$$\text{Let } x = 0.363636\dots$$

$$100x = 36.363636\dots \quad \text{two digits repeat, so } 100x$$

$$100x - x = 36.363636\dots - 0.363636\dots$$

$$99x = 36$$

$$x = 36/99 = 4/11$$

Example 2

One nine, then two.

0.666...	0.363636...
$9x = 6, \text{ so } x = 6/9$	$99x = 36, \text{ so } x = 36/99$

Example 3

The repeat does not start straight away.

Write $0.3888\dots$ as a fraction.

The 3 does not repeat. Only the 8 does.

Does $10x - x$ still work here?

Example 3

It cancels — and leaves a decimal behind.

$$x = 0.3888\ldots$$

$$10x = 3.888\ldots$$

$$10x - x = 3.888\ldots - 0.3888\ldots = 3.5$$

$$9x = 3.5, \quad \text{so} \quad x = 3.5 / 9$$

Example 3

So start one place further along.

$$10x = 3.888\ldots$$

$$100x = 38.888\ldots \quad \text{now the two tails match}$$

$$100x - 10x = 38.888\ldots - 3.888\ldots$$

$$90x = 35$$

$$x = 35/90 = 7/18$$

Whiteboards • Question 1 of 9

Boards up on three.

$$0.\overline{4}$$

Write it as a fraction.

Simplify. Show the two lines you subtracted.

Whiteboards · Question 1 of 9

Answer.

$$0.\overline{4}$$

$$10x - x = 4, \text{ so } 9x = 4$$

Answer: $\frac{4}{9}$

Whiteboards • Question 2 of 9

Boards up on three.

$$0.\overline{45}$$

Write it as a fraction.

Simplify. Show the two lines you subtracted.

Whiteboards · Question 2 of 9

Answer.

$$0.\overline{45}$$

$$100x - x = 45, \text{ so } 99x = 45$$

Answer: $\frac{5}{11}$

Whiteboards • Question 3 of 9

Boards up on three.

$$0.1\overline{6}$$

Write it as a fraction.

Simplify. Show the two lines you subtracted.

Whiteboards · Question 3 of 9

Answer.

$0.1\overline{6}$

$$100x - 10x = 15, \text{ so } 90x = 15$$

Answer: $\frac{1}{6}$

Whiteboards • Question 4 of 9

Boards up on three.

$$2.\overline{4}$$

Write it as a fraction.

Simplify. Show the two lines you subtracted.

Whiteboards · Question 4 of 9

Answer.

$$2.\overline{4}$$

the 2 comes along; $0.444... = 4/9$

Answer: $22/9$

Whiteboards • Question 5 of 9

Boards up on three.

$$0.\overline{9}$$

Write it as a fraction.

Simplify. Show the two lines you subtracted.

Whiteboards · Question 5 of 9

Answer.

$$0.\overline{9}$$

$$10x - x = 9, \text{ so } 9x = 9$$

Answer: 1

Whiteboards • Question 6 of 9

Boards up on three.

0.0 $\overline{9}$

Write it as a fraction.

Simplify. Show the two lines you subtracted.

Whiteboards · Question 6 of 9

Answer.

0.0 $\overline{9}$

one tenth of question 5

Answer: 1/10

Whiteboards • Question 7 of 9

Boards up on three.

$$x = 0.232323\dots$$

Which subtraction leaves a whole number?

$$10x - x \qquad 100x - x \qquad 100x - 10x$$

Whiteboards · Question 7 of 9

Answer.

$$x = 0.232323\dots$$

$$100x = 23.232323\dots \quad \text{and} \quad x = 0.232323\dots$$

Answer: $100x - x$, which leaves 23

Whiteboards • Question 8 of 9

Boards up on three.

$$x = 0.123123123\dots$$

Which multiplier do you need?

Give the multiplier, and the whole number it leaves.

Whiteboards · Question 8 of 9

Answer.

$$x = 0.123123123\dots$$

three digits, so $1000x = 123.123123\dots$

Answer: $1000x$, leaving 123

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Devon says $0.999\ldots$ cannot equal 1, because it is always just a little bit under.
Use today's method to settle it.

Write your answer in sentences.

This one is written work. Show the two lines you subtracted.

Whiteboards • Question 9 of 9

Answer.

Devon says $0.999\dots$ cannot equal 1, because it is always just a little bit under.
Use today's method to settle it.

$$10x = 9.999\dots, \quad \text{so } 10x - x = 9, \quad \text{so } 9x = 9$$

Answer: $x = 1$, exactly

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.